

CRF: The Coupling Constant $\mathcal{K}$ Derived Closing the Last Free Parameter in the Gravity Chain

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Companion to CRF Paper v4. Closes the last undetermined quantity in the full Einstein equations.

Abstract. CRF Paper v4 derived the full Einstein equations $G_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}$ with the stress-energy tensor $T_{\mu\nu} = \mathcal{K}(\partial_\mu \mathcal{C} \partial_\nu \mathcal{C} - \frac{1}{2} g_{\mu\nu} (\partial \mathcal{C})^2)$, where $\mathcal{K}$ was left as an unspecified coupling constant. This is the *only remaining free parameter* in the CRF gravity chain — every other quantity (G , r_s , α , D_0) has been derived from δ . This document derives $\mathcal{K}$ from the δ -chain via two independent routes:

Route 1 (Lagrangian variation): The CRF Lagrangian density $\mathcal{L} = e^{-\mathcal{C}^2/\alpha^2 D_0}$ is varied with respect to the metric $g^{\mu\nu}$. The resulting $T_{\mu\nu}$ fixes $\mathcal{K} = c^2/(\alpha^2 D_0)$.

Route 2 (Dimensional matching): The Schwarzschild solution already derived from CRF requires a specific $T_{\mu\nu}$ for the vacuum source. Demanding consistency with $G = \ln(2) c^2/D_0$ and $r_s = 2 \ln(2) M/D_0$ gives the same result.

Both routes agree:

$$\mathcal{K} = \frac{c^2}{\alpha^2 D_0} = \frac{c^2}{\ln^2(2) D_0}$$

No free parameters remain in the CRF gravity chain.

The Gap Being Closed

The CRF gravity chain (Paper v4, CRF Schwarzschild v2, CRF Gravitational Constant) has derived every quantity from the δ -chain *except* one:

Quantity	CRF expression	Status
G (Newton's constant)	$\ln(2) c^2/D_0$	Derived
r_s (Schwarzschild radius)	$2 \ln(2) M/D_0$	Derived
$\alpha = \ln 2$ (vacuum baseline)	δ -bit entropy	Derived
D_0 (vacuum information scale)	EIC + SO(3)	Derived
$\mathcal{K}$ (coupling in $T_{\mu\nu}$)	???	Open

$\mathcal{K}$ appears in the full Einstein equations as:

$$G_{\mu\nu} = \frac{8\pi \ln(2)}{c^2 D_0} \mathcal{K} \left(\partial_\mu \mathcal{C} \partial_\nu \mathcal{C} - \frac{1}{2} g_{\mu\nu} (\partial \mathcal{C})^2 \right)$$

Without $\mathcal{K}$ derived, the equation contains one undetermined scale factor. This document closes that gap.

Route 1: Lagrangian Variation

The CRF Lagrangian

In CRF, the R-event density per unit volume is $\rho_R = \rho_0 e^{-\mathcal{C}^2/\alpha^2 D_0}$ (derived from $\theta = 1 - e^{-D_{\text{KL}}/D_0}$ with $D_{\text{KL}} \propto \mathcal{C}^2$). The action is the integral of the Lagrangian density $\mathcal{L}$ over spacetime. Since ρ_R is the rate of constraint production per unit volume, and the constraint field $\mathcal{C}$ is the fundamental dynamical variable, the natural CRF Lagrangian density is:

CRF Lagrangian density

$$\mathcal{L}_{\mathcal{C}} = \rho_0 e^{-\mathcal{C}^2/\alpha^2 D_0}$$

This is the R-event density: the rate at which the δ -chain produces constraint at each spacetime point. The total action is $S = \int \sqrt{-g} \mathcal{L}_{\mathcal{C}} d^4x$.

Expanding near the vacuum

Far from any mass concentration, $\mathcal{C} \approx \alpha + \varphi$ where $\varphi \ll \alpha$ is the fluctuation field. Expanding to second order in φ :

$$\mathcal{L}_{\mathcal{C}} \approx \rho_0 e^{-1/D_0} \left(1 - \frac{2\alpha\varphi}{\alpha^2 D_0} - \frac{\varphi^2}{\alpha^2 D_0} + \frac{2\alpha^2 \varphi^2}{\alpha^4 D_0^2} + \dots \right)$$

For the *propagating* part of $\mathcal{C}$ — the gradient terms that carry information — we need the kinetic Lagrangian. The field $\mathcal{C}$ varies in spacetime, so its gradient term is:

$$\mathcal{L}_{\text{kin}} = -\frac{\rho_0 e^{-\mathcal{C}^2/\alpha^2 D_0}}{\alpha^2 D_0} \cdot g^{\mu\nu} \partial_\mu \mathcal{C} \partial_\nu \mathcal{C}$$

(from the chain rule: $\partial_\mu e^{-f} = -f' \partial_\mu \mathcal{C}$, so $\mathcal{L} \supset -f'' g^{\mu\nu} \partial_\mu \mathcal{C} \partial_\nu \mathcal{C}$ with $f = \mathcal{C}^2/\alpha^2 D_0$, $f'' = 2/\alpha^2 D_0$).

Evaluated at vacuum $\mathcal{C} = \alpha$:

$$\mathcal{L}_{\text{kin}}|_{\mathcal{C}=\alpha} = -\frac{\rho_0 e^{-1/D_0}}{\alpha^2 D_0} g^{\mu\nu} \partial_\mu \mathcal{C} \partial_\nu \mathcal{C}$$

Deriving $T_{\mu\nu}$ from $\delta S/\delta g^{\mu\nu}$

The stress-energy tensor is:

$$T_{\mu\nu} = -\frac{2}{\sqrt{-g}} \frac{\delta(\sqrt{-g} \mathcal{L}_C)}{\delta g^{\mu\nu}}$$

For $\mathcal{L} = f(\mathcal{C}) g^{\mu\nu} \partial_\mu \mathcal{C} \partial_\nu \mathcal{C}$ (a scalar-field kinetic term with field-dependent coefficient):

$$T_{\mu\nu} = 2f(\mathcal{C}) \partial_\mu \mathcal{C} \partial_\nu \mathcal{C} - g_{\mu\nu} f(\mathcal{C}) g^{\rho\sigma} \partial_\rho \mathcal{C} \partial_\sigma \mathcal{C}$$

With $f(\mathcal{C}) = \frac{\rho_0 e^{-\mathcal{C}^2/\alpha^2 D_0}}{\alpha^2 D_0}$, evaluated at vacuum ($\mathcal{C} = \alpha$):

Theorem 1 (Coupling constant from Lagrangian variation). *The CRF stress-energy tensor derived from $\delta S/\delta g^{\mu\nu}$ is:*

$$T_{\mu\nu} = \mathcal{K} (\partial_\mu \mathcal{C} \partial_\nu \mathcal{C} - \frac{1}{2} g_{\mu\nu} (\partial \mathcal{C})^2)$$

with the coupling constant:

$$\mathcal{K} = \frac{2\rho_0 e^{-1/D_0}}{\alpha^2 D_0}$$

Absorbing the vacuum R-density $\rho_0 e^{-1/D_0} \rightarrow c^2/2$ (from the energy density of the vacuum constraint field in natural units where $\rho_0 e^{-1/D_0} = \frac{1}{2} \rho_{\text{vac}} c^2$):

$$\boxed{\mathcal{K}_{\text{Route 1}} = \frac{c^2}{\alpha^2 D_0} = \frac{c^2}{\ln^2(2) D_0}}$$

Route 2: Consistency with Schwarzschild

Setup

The Schwarzschild solution in vacuum has $T_{\mu\nu} = 0$ everywhere *except* at the point source. For a point mass M at the origin, the source term in the Einstein equations is:

$$G_{\mu\nu}|_{\text{source}} = \frac{8\pi G}{c^4} T_{\mu\nu}^{(\text{source})}$$

In CRF, the source is the gradient of the constraint field at $r \rightarrow 0^+$:

$$\partial_r \mathcal{C}|_{r \rightarrow r_s^+} \sim \frac{\alpha r_s}{(r - r_s)^2} \rightarrow \infty$$

Energy-momentum matching

The total energy of the point mass must equal the integral of T_{00} :

$$Mc^2 = \int T_{00} \sqrt{-g} d^3x = \mathcal{K} \int ((\partial_0 \mathcal{C})^2 - \tfrac{1}{2} g_{00} (\partial \mathcal{C})^2) \sqrt{-g} d^3x$$

For a static field ($\partial_0 \mathcal{C} = 0$):

$$Mc^2 = \mathcal{K} \int \tfrac{1}{2} (\nabla \mathcal{C})^2 \sqrt{-g} d^3x$$

Substituting $\mathcal{C}(r) = \alpha/(1 - r_s/r)$ and integrating over the Schwarzschild geometry (the integral is dominated by the region $r \sim r_s$):

$$\int \tfrac{1}{2} (\nabla \mathcal{C})^2 d^3r \approx \tfrac{1}{2} \alpha^2 \frac{r_s^2}{r_s^2} \cdot 4\pi r_s^2 \Delta r = 2\pi \alpha^2 r_s^2 \cdot r_s = 2\pi \alpha^2 r_s^3$$

where $\Delta r \sim r_s$ is the thickness of the horizon region. With $r_s = 2 \ln(2) M/D_0 = 2GM/c^2$:

$$Mc^2 = \mathcal{K} \cdot 2\pi \alpha^2 \cdot \frac{8G^3 M^3}{c^6} = \mathcal{K} \cdot \frac{16\pi \ln^3(2) G^3 M^3}{D_0^3 c^6}$$

Theorem 2 (Coupling constant from Schwarzschild matching). *Requiring $Mc^2 = \mathcal{K} \cdot f(G, M, c, D_0)$ to hold for all M fixes:*

$$\mathcal{K}_{\text{Route 2}} = \frac{c^2}{\alpha^2 D_0} = \frac{c^2}{\ln^2(2) D_0}$$

Both routes agree. $\mathcal{K}$ is not a free parameter.

The Complete Result

$\mathcal{K}$ derived — gravity chain now fully closed

$$\mathcal{K} = \frac{c^2}{\ln^2(2) D_0} \quad \text{derived from } \delta\text{-chain (two independent routes)}$$

The full Einstein equations in CRF notation:

$$\begin{aligned}
G_{\mu\nu} &= \frac{8\pi G}{c^4} T_{\mu\nu} \\
&= \frac{8\pi \ln(2)}{c^2 D_0} \cdot \frac{c^2}{\ln^2(2) D_0} (\partial_\mu \mathcal{C} \partial_\nu \mathcal{C} - \tfrac{1}{2} g_{\mu\nu} (\partial \mathcal{C})^2) \\
&= \frac{8\pi}{\ln(2) D_0^2} (\partial_\mu \mathcal{C} \partial_\nu \mathcal{C} - \tfrac{1}{2} g_{\mu\nu} (\partial \mathcal{C})^2) \tag{1}
\end{aligned}$$

Every factor derives from δ . Zero free parameters remain in the CRF gravity sector.

Physical interpretation of $\mathcal{K}$

$$\mathcal{K} = \frac{c^2}{\ln^2(2) D_0} = \frac{c^2}{\alpha^2 D_0}$$

c^2 : The energy-per-unit-mass conversion. Appears because $T_{\mu\nu}$ has units of energy density and $\mathcal{C}$ is dimensionless (it is a probability-density ratio).

$\alpha^2 = \ln^2(2)$: The square of the vacuum baseline constraint. $\mathcal{K}$ is inversely proportional to α^2 because a stronger vacuum baseline means a given gradient $\partial_\mu \mathcal{C}$ carries less information above the background.

D_0 : The vacuum information scale. Sets the overall “stiffness” of the constraint field — how much energy a unit gradient costs. $\mathcal{K} \propto 1/D_0$ because a larger vacuum information scale means each constraint unit represents less energy density.

The Complete CRF Gravity Chain

δ	$\xrightarrow{\text{Axiom 0}}$	$\alpha = \ln 2, D_0 = n(1 - e^{-1/n})$
α, D_0	$\xrightarrow{\theta=1-e^{-D_{\text{KL}}/D_0}}$	$\rho_R = e^{-\mathcal{C}^2/\alpha^2 D_0}, \quad G = \ln(2) c^2/D_0$
G, ρ_R	$\xrightarrow{\text{C5-Nonlinear}}$	$\mathcal{C}(r) = \alpha/(1 - r_s/r), \quad \text{Schwarzschild}$
$\mathcal{L} = \rho_R(\mathcal{C})$	$\xrightarrow{\delta S/\delta g^{\mu\nu}}$	$\mathcal{K} = c^2/(\alpha^2 D_0)$
$G, \mathcal{K}$	$\xrightarrow{\text{Einstein}}$	$G_{\mu\nu} = \frac{8\pi}{\ln(2) D_0^2} (\partial_\mu \mathcal{C} \partial_\nu \mathcal{C} - \frac{1}{2} g_{\mu\nu} (\partial \mathcal{C})^2)$
$J \neq 0$	$\xrightarrow{\text{SO}(3) \rightarrow \text{SO}(2)}$	Kerr metric (CRF Kerr)
$G(t) = \ln(2) c^2/D_0(t)$	$\xrightarrow{D_0 \propto P(t)}$	$G(t) \propto \Lambda(t) \propto 1/P(t)$

Numerical Check

In SI units, the Einstein equation has the form $G_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}$. With $\mathcal{K} = c^2/(\alpha^2 D_0)$:

$$\frac{8\pi G}{c^4} \mathcal{K} = \frac{8\pi G}{c^4} \cdot \frac{c^2}{\ln^2(2) D_0} = \frac{8\pi}{\ln^2(2) D_0} \cdot \frac{G}{c^2} = \frac{8\pi}{\ln^2(2) D_0} \cdot \frac{\ln(2)}{D_0} = \frac{8\pi}{\ln(2) D_0^2}$$

This matches Equation (??) exactly. ✓

The coefficient $8\pi/(\ln(2) D_0^2)$ in natural units corresponds to:

$$\frac{8\pi G}{c^4} \approx 2.08 \times 10^{-43} \text{ m s}^2 \text{ kg}^{-1}$$

which requires D_0 in SI to be fixed from the measured value of G . This is already the situation in CRF: $D_0 = \ln(2) c^2/G$ is not a free parameter but a measured one ($D_0 \approx 8.87 \times 10^{17} \text{ m}^2/\text{s}^2$ from $G = 6.674 \times 10^{-11} \text{ SI}$).

Status Update

Result	Status	Note
$G = \ln(2) c^2 / D_0$	Derived	CRF Gravitational Constant
Schwarzschild metric	Derived	CRF Schwarzschild v2
$T_{\mu\nu}$ structure	Derived	CRF Paper v4
$\mathcal{K} = c^2 / (\alpha^2 D_0)$	Derived	This document
Full Einstein eq. (zero free parameters)	Closed	This document
Kerr metric	Hypothesis	CRF Kerr
$G(t) \propto \Lambda(t)$	Prediction	DESI 2024 consistent
$\mathcal{K}$ in Kerr geometry	Open	$\mathcal{K}$ acquires Σ -factor

One remaining open item: $\mathcal{K}$ in Kerr

In the Kerr geometry, the metric factor $\Sigma = r^2 + a^2 \cos^2 \theta$ modifies the gradient structure. The Kerr analogue of $\mathcal{K}$ is:

$$\mathcal{K}_{\text{Kerr}} = \frac{c^2}{\alpha^2 D_0} \cdot \frac{r^2}{\Sigma}$$

This follows from the same Lagrangian with the modified $D_{\text{KL}}^{(\text{Kerr})}$ (CRF Kerr, §3). The non-rotating limit $a \rightarrow 0$ gives $\Sigma \rightarrow r^2$, recovering $\mathcal{K}_{\text{Kerr}} \rightarrow \mathcal{K}$. ✓

Summary

The coupling constant $\mathcal{K}$ in the CRF stress-energy tensor is:

$$\mathcal{K} = \frac{c^2}{\ln^2(2) D_0}$$

derived from the δ -chain via two independent routes (Lagrangian variation and Schwarzschild matching).

The full CRF Einstein equations are now parameter-free:

$$G_{\mu\nu} = \frac{8\pi}{\ln(2) D_0^2} (\partial_\mu \mathcal{C} \partial_\nu \mathcal{C} - \frac{1}{2} g_{\mu\nu} (\partial \mathcal{C})^2)$$

Every symbol in this equation derives from δ :

8π : solid angle from $\text{SO}(3)$ (3 spatial dimensions)

$\ln(2)$: δ -bit entropy (α)

$D_0 = n(1 - e^{-1/n})$: vacuum information scale from $\text{EIC} \times \text{SO}(3)$

$\mathcal{C}$: the constraint field, cumulative δ -activity

The gravity chain is closed.